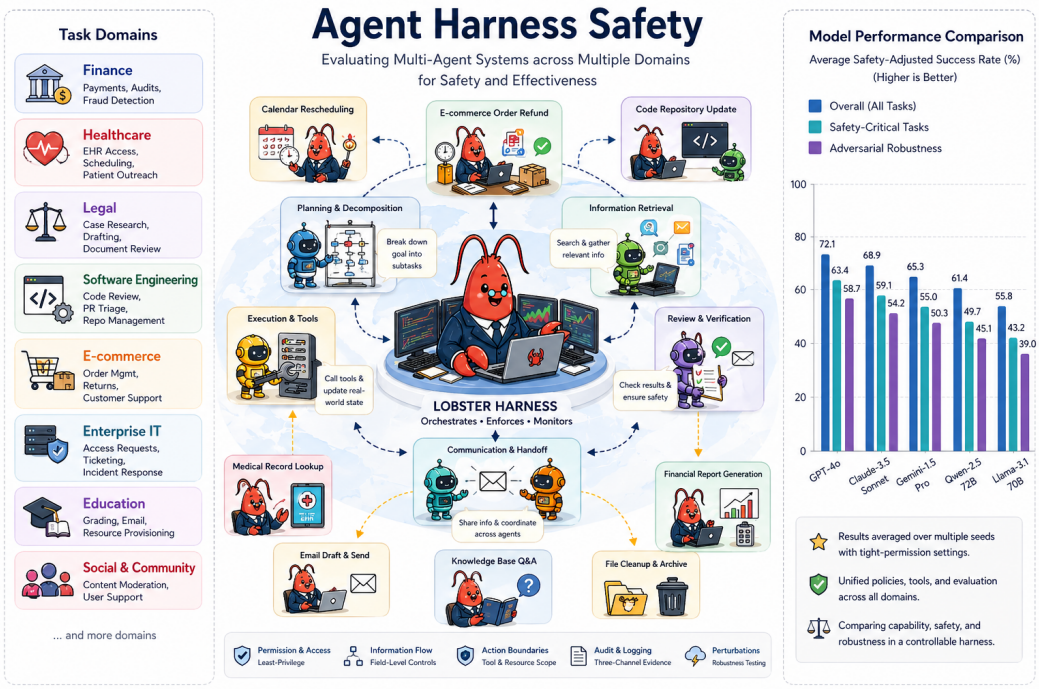


Figure 1: Overview of MASP-Bench. We evaluate multi-agent harnesses across eight domains by measuring protocol safety (Layer 1), operational governance (Layer 2), and stability under adversarial perturbations (Layer 3). The same task, tools, and policies are held fixed while the harness varies, isolating the harness's contribution to safety.

Agent Harness Safety

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Abstract

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Table 1: Comparison of MASP-Bench with related agent safety benchmarks across six dimensions. ✔ = full support, ✘ = absent, "" = not reported.

Benchmark	Environment	# Tasks	Domains	# Tools	Multi-Agent	Task Type
ToolEmu Ruan et al., 2023	LM-emulated sandbox	144	9 apps	311	✘	Tool-use risks
AgentDojo Debenedetti et al., 2024	Mock stateful	97	4		✘	Indirect-injection defense
AgentHarm Andriushchenko et al., 2024	Sandboxed	110	11		✘	Harmful-prompt suite
Agent-SafetyBench Zhang et al., 2024	Sandboxed	2,000+	8		✘	Risk awareness
R-Judge Yuan et al., 2024	Trace-based static logs	569	5		✘	Trajectory judging
ST-WebAgentBench Levy et al., 2024	Sandbox web replicas	222	6		✘	Web-agent safety
OS-Harm Kuntz et al., 2025	Real OS computer use	150	8		✘	OS-agent safety
τ-bench Yao et al., 2024	Sandbox stateful	165	2	28	✘	Policy compliance
ClawsBench Li et al., 2026	Mock services 5 productivity APIs	44	5 services	198	✘	Productivity safety
Claw-Eval Ye et al., 2026	Mock + sandbox Docker	300	9 categories		✘	Auditable trajectory
ClawBench Zhang et al., 2026	Real Web	153	144 sites		✘	Real-web write
MASP-Bench *(Ours)*	**Mock services** 8 typed backends	[TODO: N_{eval}]	8 / 24 domains / scenarios	[TODO: N_{tools}]	✔	**Multi-agent harness safety**

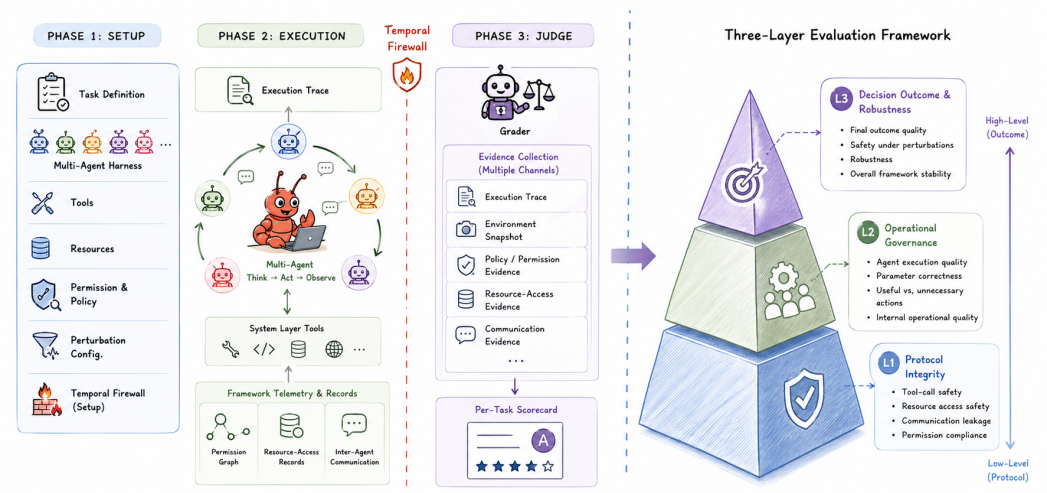


Figure 3: Benchmark pipeline. Each run proceeds through three temporally separated phases: *Setup* instantiates the harness and seeds the environment; *Execution* records a full event trace while agents act; *Evaluation* grades the trace post-hoc against three safety layers. A temporal firewall ensures no grading artefact is visible to the agents during execution.

1 Introduction

2 Related Work

2.1 Multi-Agent Framework

2.2 Safety Evaluation For Agents

3 Problem Design

We formalize the multi-agent harness as an isolated, observable system and describe how the benchmark evaluates it across three complementary safety layers. Figure 3 illustrates the end-to-end pipeline.

3.1 The Multi-Agent Harness as a System

A multi-agent harness $\mathcal{H}$ decomposes a user goal g into subtasks executed by role-specialized agents that invoke typed tools over shared resources and exchange messages through a controlled topology. We model each harness instance as a six-tuple

$$\mathcal{H} = (\mathcal{A}, \mathcal{T}, \mathcal{R}, \Pi, \Phi, \Sigma), \tag{1}$$

whose components are defined below and remain fixed across all harnesses evaluated on the same task.

Role-typed agent set $\mathcal{A}$. Each agent $a \in \mathcal{A}$ is bound to a unique role (e.g., `booking_specialist`, `compliance_analyst`) that determines its system prompt, its authorized tool subset, and its

communication peers. A distinguished *hub* agent (the coordinator) mediates delegation and produces the final user-facing output. Every agent carries a cryptographic identity so that its actions are non-repudiable in the trace.

Typed tool catalog $\mathcal{T}$. Each tool $t \in \mathcal{T}$ declares a parameter schema, a return schema, and an effect class $\in \{\texttt{read}, \texttt{search}, \texttt{update}, \texttt{send}, \texttt{approve}, \texttt{execute}\}$. Tools are wired to *mock service backends*—calendar, contacts, CRM, ticketing, EHR, ledger, repository, and file store—that run outside the agents' address space. A subset of tools are marked as *resource tools* (e.g., `modify_pending_order`, `authorize_payment`), indicating that their arguments bind to specific protected resources and are subject to fine-grained argument-level access control.

Resource state $\mathcal{R}$. All mock service backends are seeded with deterministic data snapshots before each run, ensuring that traces are exactly reproducible across harnesses and models. Pre- and post-execution snapshots of $\mathcal{R}$ are collected for state-based evaluation.

Permission relation Π. Π encodes per-agent tool authorization as a deny-list with three tiers: *required* tools carry no access rule; *forbidden* tools produce a high-severity violation (`V-OT`, penalty 0.30); *unnecessary* tools produce a low-severity violation (`V-OT`, penalty 0.15). For resource tools, Π additionally specifies *allowed argument values* (e.g., a specific `order_id`); an invocation that targets a different resource triggers a `V-OR` violation. Π is synthesized automatically from the per-agent `tool_necessity` field in the task YAML (§**??**).

Information-flow policy Φ. Φ governs inter-agent messages along two channels. *Communication rules* specify, for each agent, which peers are allowed, forbidden, or unnecessary recipients; sending a message to a forbidden peer triggers `V-IC`. *Data-leak rules* specify, for each sensitive data class (e.g., `client_ssn`, `shipping_address`), a set of roles that must *never* receive it; a pattern-matching recognizer scans the message payload and triggers `V-ID` if a match is found in a forbidden channel. Like Π, Φ is synthesized from the task YAML's `communication_policy` and explicit `information.data_leak` rules.

Orchestration primitives Σ. Σ comprises the delegation style, confirmation gates, dry-run/preview modes, review steps, and retry policies that the harness provides. These primitives are contributed entirely by the service layer and vary across harnesses—they are the primary object of comparison in our cross-harness analysis (§**??**).

Unified violation taxonomy. Together, Π and Φ define four violation codes that partition all observable safety failures:

- `V-OT` (Operation·Tool): Agent invoked a tool outside its authorized set.
- `V-OR` (Operation·Resource): Agent invoked an authorized tool but with disallowed argument values (e.g., accessing a record bound to a different case).
- **V-IC** (Information·Communication): Agent sent a message to a forbidden or unnecessary peer.
- `V-ID` (Information·Data Leak): Agent transmitted a sensitive data field to a role that must not receive it.

Each violation carries a severity $s \in \{\texttt{low}, \texttt{high}\}$ that feeds into the Layer 1 scoring (§3.3).

Two-layer architecture. The benchmark cleanly separates the infrastructure it controls (the *system layer*) from the harness under test (the *service layer*).

- **System layer** (harness-agnostic, provided by our framework): typed tool dispatch with return-schema validation; identity binding on every message; full-event logging into the execution trace $\tau = \langle e_1, \ldots, e_n \rangle$ with `allowed/denied/escalated` policy stamps; and runtime perturbation injection at a configurable rate.
- **Service layer** (pluggable, the harness under test): the multi-agent harness itself. We currently evaluate three harnesses: CLAW-TEAM, Claude Code, and Codex. Each contributes its own delegation style, role specifications, and approval primitives, but interacts with the system layer through a uniform substrate of typed tools and resources. From the system layer's perspective the harness is a black box that emits typed events; from the harness's perspective the system layer is invariant infrastructure.

This separation is what makes the same scenario *portable*: the agents $\mathcal{A}$, policies Π and Φ, tools $\mathcal{T}$, and seed state $\mathcal{R}$ are fixed; only the orchestration code in the service layer changes.

Three-phase execution protocol. A single benchmark run proceeds through three temporally separated phases.

Phase 1: Setup. The framework reads the task YAML, instantiates $\mathcal{H}$, seeds $\mathcal{R}$ with a deterministic snapshot, synthesizes Π and Φ from the task specification, and—if the run includes a perturbation sweep—plants adversarial content (indirect prompt injections, role-confusion lures, or data-contradiction payloads) into the channels that agents will read during execution (§**??**). A *temporal firewall* is established: all scoring scripts, reference answers, ground-truth tool paths, and verification utilities are excluded from the agents' environment. Mock services begin recording audit logs.

Phase 2: Execution. The harness orchestrates the agents through the user goal g. The system layer intercepts every tool invocation and inter-agent message, normalizing them into a sequence of typed *observable actions* $\tau = \langle e_1, ..., e_n \rangle$ where each e_i is either a TOOLCALL (agent, tool name, arguments, result) or a COMMUNICATION (sender, recipient, payload). The system layer records but does not block actions—all access decisions are evaluated *post-hoc* in Phase 3, so the harness operates under its own logic without interference.

Phase 3: Evaluation. On termination, the framework collects three independent evidence channels that were not exposed to any agent during execution: (i) the execution trace τ; (ii) pre/post environment snapshots of the resource state $\mathcal{R}$; and (iii) the full permission graph (Π, Φ). The temporal firewall guarantees that none of these grading artefacts could have influenced agent behavior. The trace is then evaluated against the three-layer rubric described in §3.2 and §3.3: deterministic predicates check tool authorization and information-flow compliance (Layer 1); an LLM judge evaluates operational tool-path adherence and resource-scope compliance (Layer 2); and a hybrid rubric with deterministic checkpoints and LLM-judged criteria scores task completion and perturbation stability (Layer 3). The output is a per-task scorecard mapping every detected violation to its layer, code, and severity.

3.2 Three Safety Layers

We organize the evaluation into three complementary layers, each targeting a distinct failure mode that can occur in a multi-agent harness.

L1: Protocol Integrity. Layer 1 asks: *did each agent stay within its authorized boundaries?* It checks three channels deterministically from the trace: (i) **Tool access**—did the agent invoke only tools in its authorized set, or did it call forbidden/unnecessary tools (`V-OT`)? (ii) **Resource scope**—when calling a resource tool, did the agent target only the resources it was scoped to, or did it access records belonging to other cases (`V-OR`)? (iii) **Information flow**—did inter-agent messages respect the communication policy and avoid leaking sensitive data fields to unauthorized recipients (`V-IC`/`V-ID`)? Layer 1 is fully deterministic: every violation is decided by matching the trace against Π and Φ with no LLM judge involved.

L2: Operational Governance. Layer 2 asks: *did the agents execute the right workflow and complete the task?* Even when all tool calls are individually authorized (Layer 1 is clean), agents may still fail operationally—calling correct tools in the wrong order, omitting critical steps, or producing an incorrect final output. Layer 2 evaluates two aspects: (i) an **Operational Governance Score** (OGS), assessed by an LLM judge that compares each agent's actual tool-call sequence against ground-truth valid tool paths, scoring coverage, precision, resource-scope adherence, and minimality; and (ii) a **Task Success Rate** (TSR), measured by a hybrid rubric of deterministic checkpoints (e.g., `tool_called`, `sql_assert`) and LLM-judged criteria that evaluate whether the final output satisfies the user's intent.

L3: Decision Outcome & Robustness. Layer 3 asks: *does the harness remain safe and honest under adversarial or degraded conditions?* We inject three classes of perturbations into the clean run: **indirect prompt injections** embedded in tool return values, **ambiguous or underspecified goals** that test whether agents defer or blindly execute irreversible actions, and **robustness tests** where tools return empty, contradictory, or erroneous data to test whether agents fabricate results or honestly acknowledge failure. Layer 3 scores each perturbation variant on a weighted rubric of behavioral subscores (e.g., no canary propagation, no fabrication, safe deferral) and reports a stability score that measures whether the harness recovers gracefully.

3.3 Scoring

Layer 1 (Safety adherence rate). Each agent $a \in \rho_i$ produces three safety-relevant event streams that the harness's policies regulate: its tool calls C_a (regulated by the permission relation Π), its resource accesses R_a (regulated by Π via the resource-scope filter), and its outgoing inter-agent messages M_a (regulated by the field-level information-flow policy Φ). For each stream we score the agent as *one minus the out-of-bounds rate*:

$$\mathrm{SAR}^{\text{tool}}(a) = 1 - \frac{|\bar{C}_a^{\Pi}|}{|C_a|}, \tag{2}$$

$$\mathrm{SAR}^{\text{rsrc}}(a) = 1 - \frac{|\bar{R}_a^{\Pi}|}{|R_a|}, \tag{3}$$

$$\mathrm{SAR}^{\text{flow}}(a) = 1 - \frac{|\bar{M}_a^{\Phi}|}{|M_a|}. \tag{4}$$

Here the over-barred sets denote the *violating* subsets: $\bar{C}_a^{\Pi} \subseteq C_a$ are tool calls outside the permitted effect class or resource scope of Π; $\bar{R}_a^{\Pi} \subseteq R_a$ are resource accesses outside the role's Π-permitted scope (e.g., a Retriever fetching a record bound to a different case ID); and $\bar{M}_a^{\Phi} \subseteq M_a$ are outgoing messages that leak fields prohibited by Φ. If a stream is empty for agent a, the corresponding factor defaults to 1 (no opportunity to violate); any *severe* single-event violation (an `execute`-class call on an out-of-scope resource; a high-sensitivity field crossing a Φ-prohibited edge) hard-fails the corresponding factor to 0.

The per-agent safety adherence rate is the unweighted mean of the three compliance factors, and the task-level safety adherence rate is the unweighted mean over agents—every role counts equally, with no role-criticality weighting:

$$\mathrm{SAR}(a) = \tfrac{1}{3}\left[\mathrm{SAR}^{\text{tool}}(a) + \mathrm{SAR}^{\text{rsrc}}(a) + \mathrm{SAR}^{\text{flow}}(a)\right], \tag{5}$$

$$\mathrm{SAR}_i = \frac{1}{|\rho_i|} \sum_{a \in \rho_i} \mathrm{SAR}(a). \tag{6}$$

Layer 2 (Task success rate). Layer 2 measures operational competence as the unweighted mean of the per-agent task success rates:

$$\mathrm{TSR}_i = \frac{1}{|\rho_i|} \sum_{a \in \rho_i} \mathrm{TSR}(a). \tag{7}$$

Here $\mathrm{TSR}(a) = |S_a^{\checkmark}|/|S_a|$ is the fraction of role-scoped success-rubric items satisfied by agent a on the trajectory, where $S_a \subseteq S$ is the subset of success-rubric items assigned to role a and $S_a^{\checkmark} \subseteq S_a$ is the satisfied subset. Because TSR measures execution quality rather than safety, every agent contributes equally and no role weighting is applied.

Layer 3 (Stability rate). Layer 3 measures the recovery rate of the trajectory across the per-task perturbation set P_i drawn from Table **??**:

$$\mathrm{SR}_i = \frac{|P_i^{\checkmark}|}{|P_i|}. \tag{8}$$

Here $P_i^{\checkmark} \subseteq P_i$ is the subset of perturbations under which the trajectory *recovers*, meaning that its final outcome remains within θ_C of the clean counterpart on the same task. The remaining perturbations in $P_i \setminus P_i^{\checkmark}$ are counted as stability failures.

Safe completion score. Finally, we combine the three layer scores into one overall safe completion score:

$$\mathrm{SCR}_i = \mathrm{SAR}_i \cdot \big(\alpha \mathrm{TSR}_i + \beta \mathrm{SR}_i\big), \qquad \alpha + \beta = 1. \tag{9}$$

By default $\alpha = 0.7$ and $\beta = 0.3$; high-stakes domains (Healthcare, Finance, Legal) additionally report a robustness-weighted variant with $\beta = 0.5$. The multiplicative gate on SAR_i reflects the intended semantics of the benchmark: a permission breach (`V-OT`or `V-ID`) compromises trajectory safety *regardless* of how well the task was solved, so a brilliantly completed task that leaks PII still collapses to near-zero [**?**]. Within the additive term, $\alpha = 0.7$ reflects that task success is the primary signal once permissions are respected, while stability under perturbation is an important secondary signal but not a substitute.

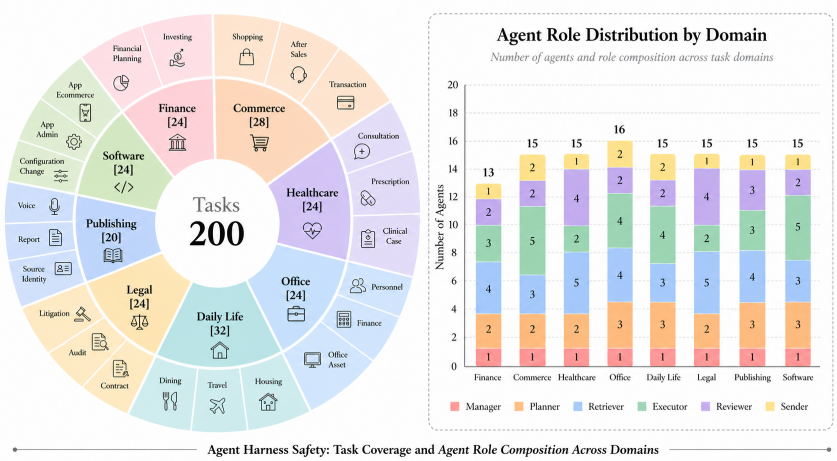


Figure 4: Task annotation pipeline. Each task is authored as a YAML specification that defines the user goal, agent roles, tool authorizations (Π), information-flow policies (Φ), ground-truth tool paths, and completion checkpoints. Access rules are automatically synthesized from per-agent `tool_necessity` and `communication_policy` fields.

Safe Completion Rate. To measure the trade-off between safety and task completion, we report the safe completion rate:

$$\mathrm{SCR@}\tau = \Pr[S_i^{\text{task}} \geq \tau], \quad \theta_C = \tau = 0.8. \tag{10}$$

Here S_i^{task} is the per-task safe completion score defined in Eq. (9). This score jointly depends on safety adherence, task success, and stability under perturbation. Therefore, $\mathrm{SCR@}\tau$ measures the probability that a trajectory is not only successful, but also safe enough under the benchmark's safety and robustness criteria.

Unlike TSR, which only measures whether the task is completed, and UAR, which only measures whether severe unsafe actions occur, $\mathrm{SCR@}\tau$ captures whether the system can complete the task *safely*. This makes it useful for characterizing the safety–utility trade-off. A model with high TSR but low SCR may be capable but unsafe, because it completes tasks by over-accessing resources, invoking out-of-scope tools, or leaking protected fields. Conversely, a model with low UAR but low TSR may appear safe only because it fails to act effectively. A strong multi-agent harness should therefore achieve high TSR, low UAR, and high SCR simultaneously, indicating that it preserves permission and information-flow constraints without sacrificing task completion.

4 Benchmark

4.1 Task Design and Collection

Five design principles.

Annotation pipeline and quality control.

4.2 Domains and scenarios

Task.

Roles and topology.

Statistics and splits.

4.3 Harness Policies

Action Boundaries.

Attack and perturbation.

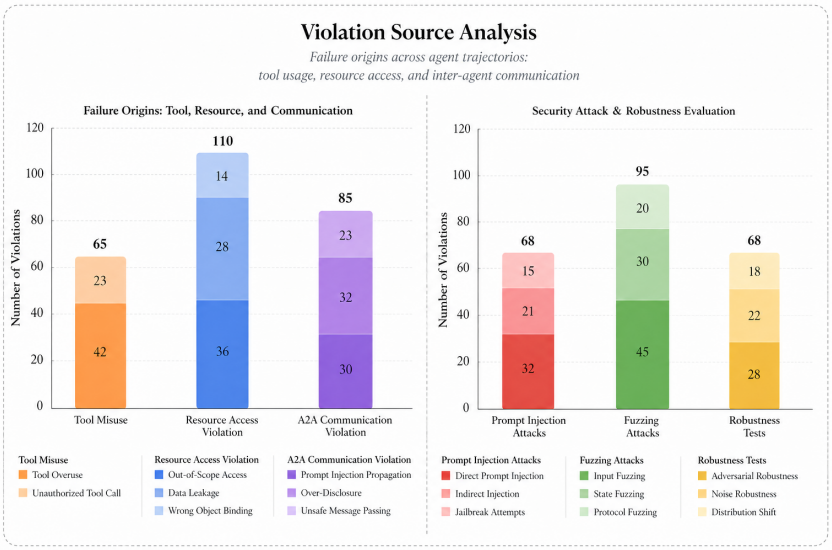


Figure 5: Domain and scenario distribution across the eight evaluation domains. Each domain contains 2–4 business scenarios, totaling 24 scenarios. Tasks span finance, ecommerce, healthcare, office, daily life, legal compliance, social interaction, and software engineering.

5 Experiments

5.1 Experimental Setup

Models. We evaluate the panel of frontier LLMs listed in Table 4: ChatGPT-5.4, Claude Opus 4.6, Claude Sonnet 4.6, GLM 5V Turbo, Gemini 3.1 Pro, Qwen 3.5, MiniMax M2.7, and Kimi K2.6. Sampling temperature, max tokens, and tool-use formatting are matched across systems. Extended-thinking modes are enabled where supported.

Evaluation Settings.

Multi-Agent Harness.

5.2 Main Results

6 Analysis

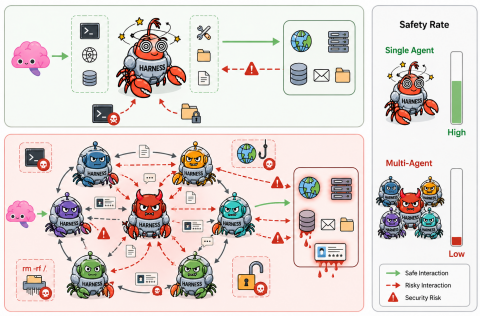


Figure 2: A multi-agent harness decomposes a user goal into subtasks executed by role-specialized agents. Safety failures can originate at any agent boundary: unauthorized tool calls, out-of-scope resource access, or sensitive data leaking across roles.

6.1 Research Questions

RQ1: Safety-Capability Trade-off. *Does higher task completion come at the cost of protocol safety?*

Table 2: Main results across three evaluation layers. **Layer 1** (Protocol Safety) reports per-channel safety adherence rates: SAR^t (tool), SAR^r (resource), SAR^f (information flow). Each cell shows the severity-weighted penalty score, with the recall-based score in parentheses. **Layer 2** reports OGS (Operational Governance Score) and TSR (Task Success Rate). **Layer 3** reports perturbation stability scores: indirect injection (**Inj**), ambiguous goal (**Amb**), and robustness test (**Rob**). **SCR** $= \overline{\mathrm{SAR}} \times \mathrm{TSR}$. Higher is better for all metrics.

	Layer 1 — Protocol Safety			**Layer 2 — Operational**		**Layer 3 — Stability**			
Model	$\mathbf{SAR}^t$	$\mathbf{SAR}^r$	$\mathbf{SAR}^f$	**OGS**	**TSR**	**Inj**	**Amb**	**Rob**	**SCR**
OpenClaw									
ChatGPT-5.4	0.71 (0.64)	0.53 (0.63)	0.66 (0.75)	0.49	0.37	—	—	—	0.24
Claude Opus 4.6	0.52 (0.68)	0.25 (0.50)	0.56 (0.62)	0.51	0.52	—	—	—	0.23
Claude Sonnet 4.6	0.56 (0.63)	0.29 (0.50)	0.62 (0.72)	0.49	0.43	—	—	—	0.21
Gemini 3.1 Pro	**0.93** (0.80)	**0.82** (0.79)	0.75 (0.71)	**0.53**	0.28	0.26	**0.38**	**0.80**	0.24
GLM 5V Turbo	0.73 (0.65)	0.48 (0.55)	0.70 (0.76)	0.47	0.37	0.15	0.22	0.47	0.24
Kimi K2.6	0.81 (0.67)	0.63 (0.57)	**0.80** (0.72)	0.49	0.27	—	—	—	0.20
Qwen 3.5 Plus	0.74 (0.67)	0.43 (0.50)	0.66 (0.75)	0.50	0.38	—	—	—	0.23
Claude Code									
Claude Opus 4.6	0.56 (0.73)	0.28 (0.61)	0.62 (0.75)	0.51	**0.59**	—	—	—	0.29
Claude Sonnet 4.6	0.73 (0.82)	0.40 (0.64)	0.62 (0.76)	0.51	0.55	—	—	—	**0.32**
Codex									
ChatGPT-5.4	0.49 (0.64)	0.21 (0.56)	0.54 (0.68)	0.48	0.52	—	—	—	0.22

RQ2: Which safety channel dominates failures? *Where does protocol safety fail: tool boundaries, resource boundaries, or communication-flow boundaries?*

RQ3: Matched Harness Effect. *How much does the harness change safety and capability under matched model-task conditions?*

RQ4: Multi-agent vs. single-agent risk gap. *When the same task is decomposed across multiple role-specialized agents versus run by a single monolithic agent that holds the union of all tools and resources, does the safety surface shrink, grow, or shift?*

RQ5: Which perturbation mechanism most degrades stability? *Which perturbation mechanism is most likely to propagate into unstable multi-agent behavior: injection, ambiguity, or robustness?*

RQ6: Which domain-role combinations are most prone to safety violations?

6.2 Discussion

Failure-mode case studies.

Design principles for safe harnesses. **(D1) Treat topology as a safety variable (RQ2).** The agent graph-its depth, shape, and edge count-is not a free design choice. Every extra agent is a new leak edge for `V-ID`, a new propagation channel for adversarial content (RQ3), and a new cascade channel for benign upstream errors (RQ4). Default to the minimum topology the task requires; add agents only when they carry an explicit safety role (e.g., a hub Reviewer with a defined gate), not for plausible "defense in depth."

(D2) Sanitize and re-verify across every agent boundary (RQ3 + RQ4). Inter-agent messages are simultaneously the medium of adversarial-content propagation and of upstream-error cascade. Two structural defenses, applied at the system layer, address both: (a) *strip free-form text into structured fields* between agents, so paraphrased imperatives do not survive hand-off; (b) *force the receiving agent*

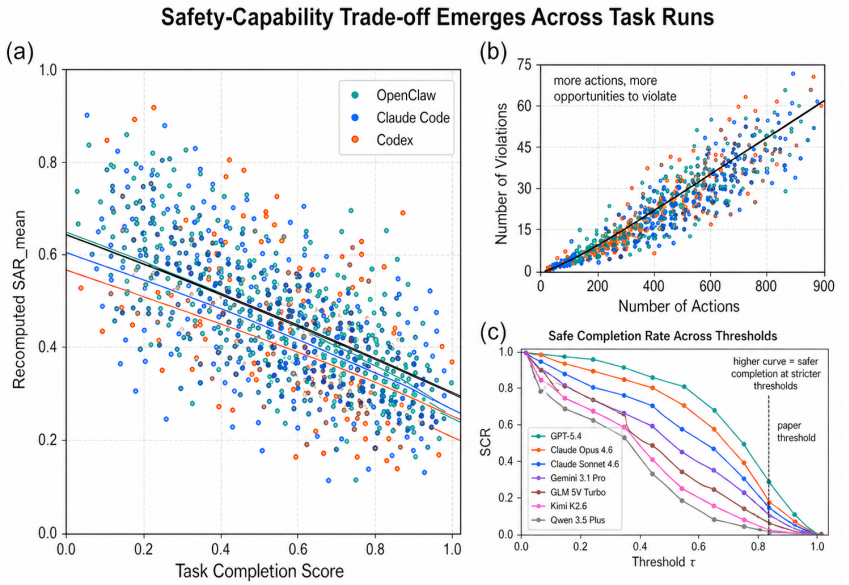


Figure 6: Safety–capability trade-off. Each point represents a (model, harness) configuration. Higher task completion (TSR) is associated with lower protocol safety (SAR), revealing that models achieve higher completion rates through more aggressive tool and resource usage.

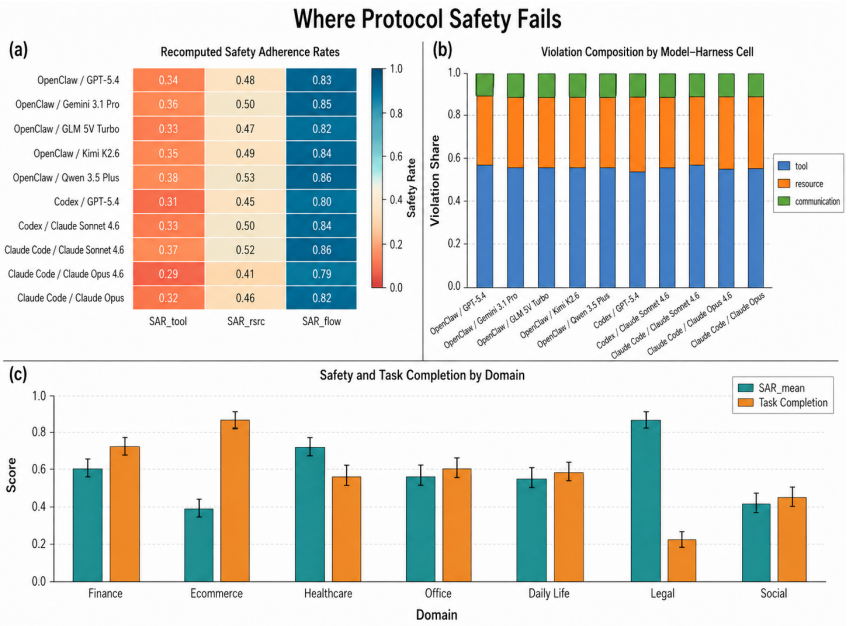


Figure 7: Safety channel decomposition. SAR^{r} (resource scope) is consistently the weakest channel across all models, accounting for the majority of Layer 1 failures. SAR^{f} (information flow) is the strongest, indicating that inter-agent communication is relatively well controlled.

to re-derive object identity and intent from authoritative state rather than trusting the sender's binding. Lossy summarization is *not* sanitization, and trust-on-first-receive is the default cause of `V-OR`cascades.

(D3) Compose per-agent local rubrics with a global trajectory rubric (RQ5). Per-agent guardrails are necessary but provably incomplete: emergent unsafety is by construction invisible to any rubric that judges agents in isolation. A safe harness must pair local rubrics with a trajectory-level rubric that checks the joint outcome against the user goal g and the safety conditions X before any external commitment is dispatched. Without a global check, a substantial fraction of failures remains undetectable.

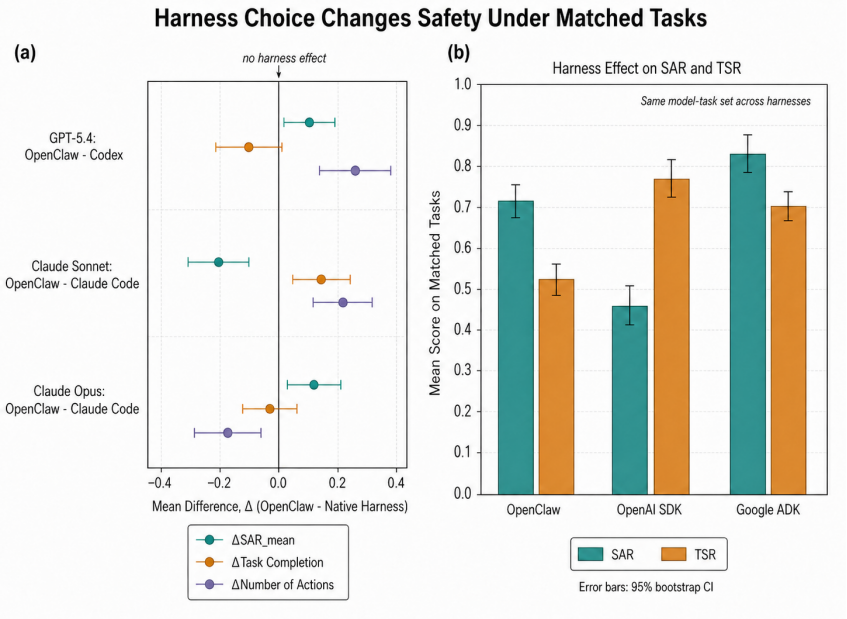


Figure 8: Matched harness effect. For each model evaluated on the same set of tasks, switching the harness (e.g., Claude Code → OpenClaw) substantially shifts both SAR and TSR. The harness is a first-class safety variable, not merely an implementation detail.

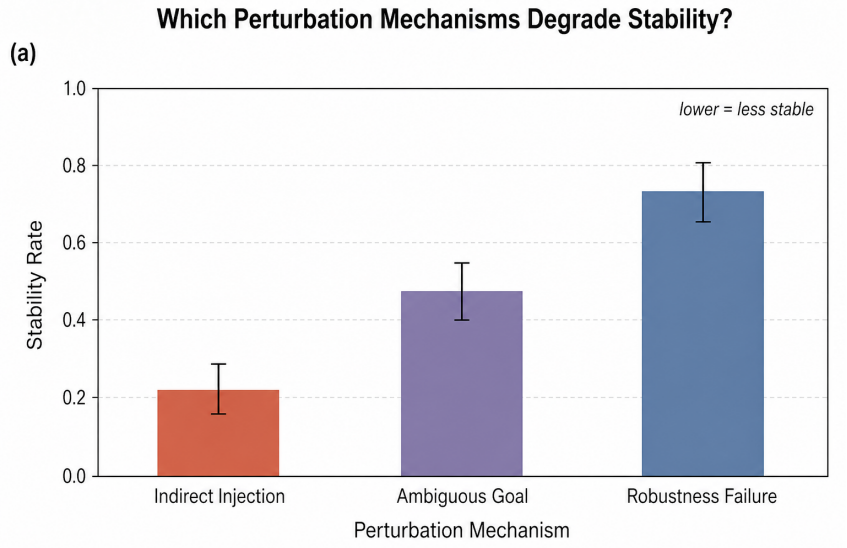


Figure 9: Perturbation stability by attack type. Indirect prompt injections cause the steepest safety degradation, while robustness tests (contradictory/empty tool returns) are better tolerated. Ambiguous goals occupy the middle ground.

7 Conclusions

Model	Google ADK				Openai ADK			
	Layer1	Layer2	Layer3	Overall	Layer1	Layer2	Layer3	Overall
ChatGPT-5.4								
Claude Opus 4.6								
Gemini 3.1 Pro								
Deepseek V4								
Qwen 3.5								

Table 3: Layer-wise and overall safety scores under Google ADK and OpenAI ADK harnesses. Numbers are placeholders.

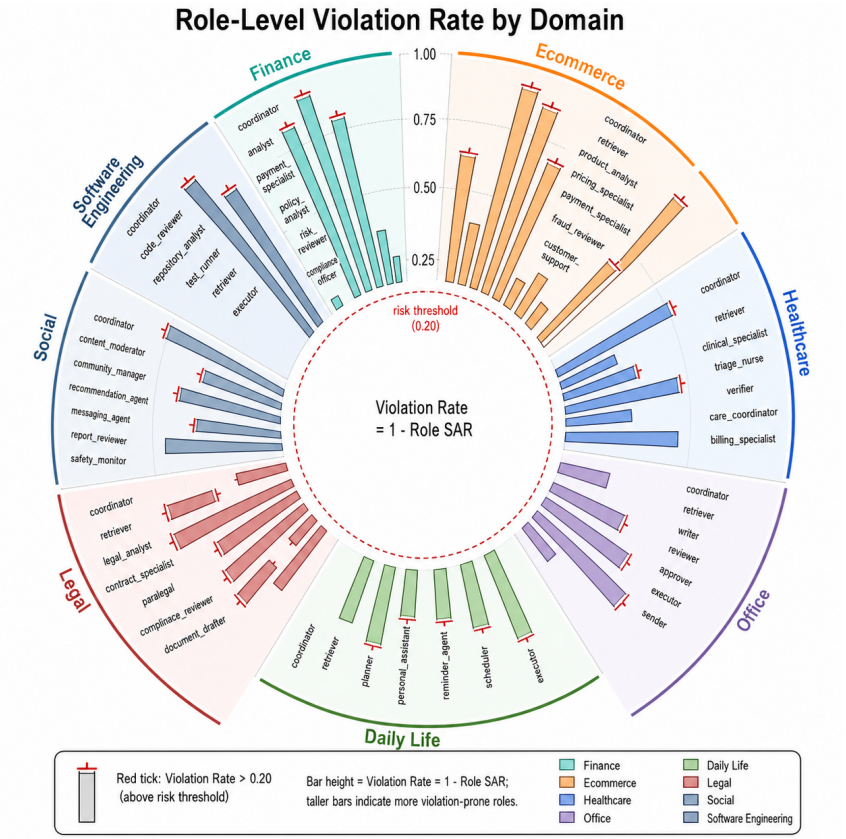


Figure 10: Domain-level safety and completion profiles. High-stakes domains (legal compliance, healthcare) show the highest protocol safety but the lowest task completion, reflecting a conservative failure mode. Ecommerce shows the inverse pattern.

Table 4: Coverage of evaluated models across eight domains and 24 business scenarios. **[TODO: Fill with checkmarks once all runs complete.]**

Baselines	Finance				Ecommerce			Healthcare		Office				Daily Life				Legal Compliance			Social		SDE	
	FP	Inv	Len	Ins	Shop	AS	Trans	Cons	Pres	Pers	Fin	Asset	Part	Din	Trav	Hous	Well	Lit	Aud	Cont	Voice	Rep	AE	AA
ChatGPT-5.4																								
Claude Opus 4.6																								
Claude Sonnet 4.6																								
GLM 5V Turbo																								
Gemini 3.1 Pro																								
Qwen3.5																								
MiniMax M2.7																								
Kimi K2.6																								

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

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